

Lecture 4: Memory & Storage

Learning Outcomes

- Recall the Memory Unit from Lecture 3's block diagram, and distinguish clearly between memory, local storage, and cloud storage.
- Explain RAM, its key properties, and the basic difference between SRAM and DRAM.
- Explain ROM, its purpose, and the main types of ROM.
- Explain cache memory and why it is needed between the CPU and RAM.
- Explain how an SSD works, and compare it with a traditional hard disk.
- Explain how a USB flash drive works and list its common uses.
- Define cloud storage, explain how it works, and list its advantages and limitations.
- Choose the right storage option — RAM, ROM, cache, SSD, USB, or cloud — for a given real-world need.
- Answer short, medium, and long-answer (up to 15 marks) examination questions on this topic with full explanations, diagrams, and examples.

1 Recap — From the Block Diagram to Memory & Storage

In Lecture 3, the Memory Unit appeared as just one box in the block diagram, briefly split into Primary Memory (fast, closely coupled to the CPU) and Secondary Memory (external, larger, permanent). This lecture zooms into that single box and studies each memory and storage technology inside it in detail — and adds one modern idea that goes beyond the traditional block diagram entirely: cloud storage, where your data can live outside your device altogether.

Important Point for Exams

Keep Lecture 3's distinction in mind throughout this lecture: RAM, ROM, and Cache are all forms of memory used directly by the CPU; SSDs and USB drives are secondary storage; and cloud storage is a newer category that behaves like secondary storage but lives on someone else's server, accessed over the internet.

2 Overview: Memory, Storage, and the Cloud



MEMORY — Short-Term & Working

Memory holds the instructions and data a program needs right now. It is extremely fast so the CPU is never kept waiting, but it is volatile — the moment power is lost, everything it holds disappears. Think of it as the space actively in use while you work.



STORAGE — Long-Term & Permanent

Secondary storage holds files, programs, and the operating system permanently. It is non-volatile — data survives even with the power off — but it is much slower to access than memory. Every photo, document, and installed app you own lives here.

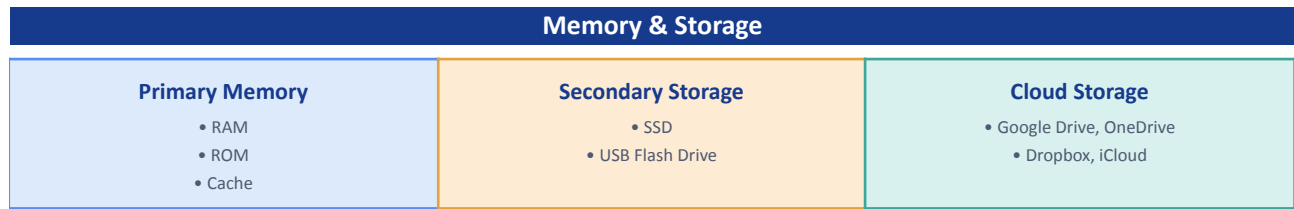


Fig 4.1 — The three broad categories covered in this lecture.

Category	Where the Data Lives	Typical Speed	Survives Power Off?
Primary Memory (RAM, ROM, Cache)	Inside or very close to the CPU	Fastest	RAM: No. ROM: Yes.
Secondary Storage (SSD, USB)	Inside or plugged into your own device	Slower than memory, still fast for SSD	Yes — non-volatile
Cloud Storage	On a remote server, accessed over the internet	Depends on your internet connection	Yes — stored elsewhere, independent of your device

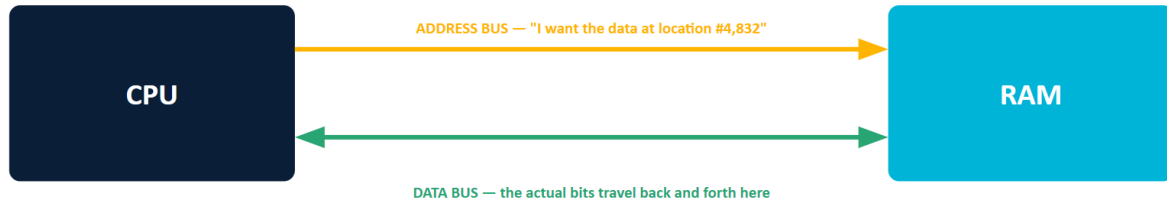
3 RAM — Random Access Memory

RAM is the computer's main working memory. Whenever a program — a browser, a game, a word processor — is opened, the operating system copies its instructions and current data from storage into RAM, because the CPU can read and write RAM thousands of times faster than it can read a disk.

Property	Explanation
Very Fast	Typical access time is measured in nanoseconds — billions of operations per second.
Randomly Addressable	Any memory cell can be reached directly using its address, in constant time — unlike a tape that must be read in order.
Volatile	All contents are lost the instant power is removed, which is why unsaved work disappears after a crash or shutdown.

How the CPU Talks to RAM

Every access happens over two separate sets of wires: the address bus and the data bus



- 1 The CPU places the memory address it needs on the address bus.
- 2 RAM decodes that address and locates the exact row and column of memory cells.
- 3 Data flows along the data bus — into the CPU on a read, or out of it on a write.
- 4 Because addressing is direct, this whole exchange takes only tens of nanoseconds, regardless of where in RAM the data is stored.

Two Kinds of RAM — SRAM and DRAM

RAM chips are broadly built using one of two technologies.

Point	DRAM (Dynamic RAM)	SRAM (Static RAM)
Used for	Main system RAM	CPU cache
Speed	Fast	Even faster
Needs refreshing?	Yes, thousands of times per second	No
Cost & density	Cheaper, higher capacity per chip	More expensive, lower capacity per chip
Important Point for Exams Both DRAM and SRAM are volatile — both lose their contents completely when power is switched off. "Volatile" describes RAM in general, not just one type of it.		

DRAM — Dynamic RAM, In Detail

The technology used for the main memory sticks inside almost every PC, laptop and phone



Each bit of DRAM is stored as an electrical charge in a tiny capacitor, paired with one transistor that acts as a switch to read or write that charge. This "1 transistor + 1 capacitor" (1T1C) design is extremely small, which is why DRAM chips can pack billions of bits into a tiny area — making it cheap enough for large capacities.

Why It Needs "Refreshing"

A capacitor slowly leaks its charge, like a bucket with a tiny hole — within milliseconds the stored bit would fade and be lost. To prevent this, the memory controller reads and rewrites (refreshes) every single cell thousands of times per second. This refresh cycle uses power and briefly blocks normal access, which is part of why DRAM is slower than SRAM.

DRAM Generations (DDR)

- DDR3** ~1600 MT/s, low power, older laptops/desktops
- DDR4** ~3200 MT/s, most common in 2015–2022 PCs
- DDR5** ~4800+ MT/s, current generation, higher bandwidth

"DDR" = Double Data Rate: data transfers on both the rising and falling edge of the clock signal, doubling throughput.

SRAM — Static RAM, In Detail

The technology used for CPU cache and registers, where every nanosecond counts



Each bit of SRAM is stored in a small circuit called a flip-flop, typically built from six transistors. A flip-flop is bistable — it naturally settles into one of two stable states (0 or 1) and stays there as long as power is supplied, with no leakage to compensate for. This makes SRAM inherently faster and more reliable, but also much larger and more expensive per bit than DRAM.

No Refresh Needed

Because the flip-flop holds its state actively, there is no leakage to correct — SRAM is ready to be accessed on every single clock cycle.

Much Faster

Access times are often under 1–2 nanoseconds — fast enough to keep up with a modern CPU core running at several gigahertz.

Larger & Costlier

Six transistors per bit (versus DRAM's two components) means far fewer bits fit in the same silicon area, so SRAM is reserved for small, critical memory like cache.

4 ROM — Read Only Memory

ROM is a non-volatile memory chip — its contents survive completely intact even with the power off. It is used to store fixed instructions the computer needs immediately at start-up, before it can even load an operating system from disk, such as the BIOS or UEFI firmware, as well as firmware in devices like routers, printers, and washing machines.

Point	RAM	ROM
Volatility	Volatile — loses data when powered off	Non-volatile — keeps data with no power at all
Read/Write	Read AND write — changes constantly during use	Read-only, or very hard to change

Point	RAM	ROM
What it stores	Currently running programs and data	Permanent start-up and firmware code
Typical capacity	Large, measured in gigabytes	Small, measured in kilobytes to megabytes

Types of ROM

ROM is not a single technology — it is a small family, and each type differs in how (or whether) it can be reprogrammed.

Type	Can It Be Changed?
Mask ROM	No — set permanently into the chip during manufacturing.
PROM (Programmable ROM)	Written once by the user with a special “burner” device; cannot be changed afterwards.
EPROM (Erasable PROM)	Can be erased using ultraviolet (UV) light and reprogrammed — but the whole chip must be erased at once.
EEPROM (Electrically Erasable PROM)	Can be erased and rewritten electrically, without UV light, often one byte at a time.

Important Point for Exams

Notice the trend: each newer type of ROM made stored data progressively easier to change — from “impossible after manufacturing” (Mask ROM), to “a one-time factory-like write” (PROM), to “UV light and several minutes” (EPROM), to finally “an electrical pulse in milliseconds” (EEPROM). Flash memory, used in SSDs and USB drives, is a direct descendant of EEPROM.

Mask ROM & PROM

The two earliest, least flexible members of the ROM family



Mask ROM

- The data pattern is built directly into the chip's physical circuitry during manufacturing, using a photographic “mask”.
- Extremely cheap to produce in bulk, but only economical for very large orders of the exact same chip.
- Cannot be changed after manufacturing — a single design mistake means the whole batch is scrapped.
- Historically used in simple calculators, toys, and early game cartridges.



PROM

(Programmable Read-Only Memory)

- Sold blank, then written once by the customer using a special device called a “PROM programmer” or “burner”.
- Inside, each bit is a tiny fuse. Programming sends a pulse of current that permanently blows selected fuses — an intact fuse reads as one value, a blown fuse reads as the other.
- Once fuses are blown, the change is irreversible — there is no way to “unblow” a fuse.
- Useful for small production runs where mask ROM's setup cost is not worth it.

EPROM — Erasable PROM

The first ROM you could wipe clean and reuse



EPROM stores each bit in a "floating-gate transistor" — a transistor with an extra, electrically isolated gate that can trap electrons. Once electrons are trapped there during programming, they stay put for years because the floating gate is surrounded by an insulator, with no path to leak away — this is what gives EPROM its non-volatility.

How It's Erased

Older EPROM chips have a small quartz window on top of the package. Shining strong ultraviolet (UV) light through this window for 10–20 minutes gives the trapped electrons enough energy to escape the floating gate, resetting every cell to its default value at once.

Whole-Chip Erase Only

There is no way to erase a single byte — UV light resets the entire chip together, so any reprogramming means erasing and rewriting everything from scratch.

Practical Limitations

The erase-and-reprogram cycle is slow and requires removing the chip from the circuit and placing it under a UV lamp, which made EPROM impractical for frequent updates.

EEPROM — Electrically Erasable PROM

The ROM that can be reprogrammed without ever leaving the circuit



EEPROM uses the same floating-gate principle as EPROM, but adds a thinner insulating layer that allows electrons to tunnel back out under an applied electric field — a phenomenon called Fowler-Nordheim tunneling. This means erasing no longer needs UV light: it can be done electrically, right inside the running device, and often one byte at a time instead of the whole chip at once.

Byte-Level Access

Individual bytes can be erased and rewritten without disturbing the rest of the chip — ideal for storing small settings that change occasionally, like a router's configuration.

Limited Write Cycles

Each erase-write cycle slightly wears the insulating layer. Typical EEPROM chips are rated for around 100,000 to 1,000,000 write cycles per cell before they may become unreliable.

The Ancestor of Flash

Flash memory (used in SSDs, USB drives, and SD cards) is essentially EEPROM redesigned to erase in large blocks at once, trading fine-grained control for much higher speed and density.

5 Cache Memory

Modern CPUs can execute an instruction in a fraction of a nanosecond, but fetching data from RAM takes tens of nanoseconds — a mismatch that would leave the CPU idle much of the time. Cache is a small pool of very fast memory sitting between the CPU and RAM that keeps a copy of recently and frequently used data close by, so the CPU rarely has to wait for slower RAM.

The Principle of Locality — Why Caching Actually Works

Temporal Locality

If a piece of data was used recently, it is very likely to be used again soon — for example, a loop counter variable that is read and updated on every pass through a loop.

Spatial Locality

If one memory address was used, nearby addresses are also likely to be used soon — for example, the next few instructions in a program, or adjacent elements of an array.

Level	Position	Typical Speed
L1	Smallest, built directly into each CPU core	Fastest
L2	Larger, still close to each core	Very fast
L3	Largest, shared across all CPU cores	Fast, but slower than L1/L2

Cache Hit vs Cache Miss

Every memory request ends in one of two outcomes

CACHE HIT

The data the CPU needs is already sitting in cache. It is delivered in just a few nanoseconds — the fastest possible outcome, keeping the CPU running at full speed with no waiting.

Typical hit rate: 90–97%

CACHE MISS

The needed data is not in cache. The CPU must wait while it is fetched from the slower RAM — tens of nanoseconds instead of a few — and that data (plus its neighbors, thanks to spatial locality) is then copied into cache for next time.

Costs 10–20x longer than a hit

Why hit rate matters so much:

Even a small drop in hit rate has an outsized effect on performance, because a miss costs 10–20 times longer than a hit. This is why CPU designers spend enormous effort on cache size, associativity, and prefetching algorithms that predict what data will be needed next.

6 RAM vs ROM vs Cache — Quick Comparison

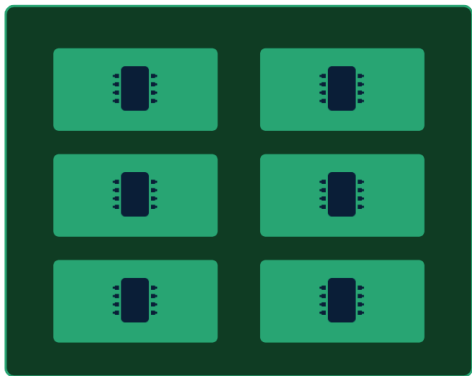
Point	RAM	ROM	Cache
Volatile?	Yes	No	Yes
Speed	Fast	Slower than RAM	Fastest of the three
Read/Write	Both	Read-only (mostly)	Both
Main purpose	Runs currently active programs	Stores permanent start-up firmware	Speeds up CPU access to frequently used data
Typical size	Gigabytes	Kilobytes to megabytes	Kilobytes to a few megabytes

7 SSD — Solid State Drive

A Solid State Drive (SSD) is a secondary storage device built entirely from NAND flash memory chips — the same underlying technology descended from EEPROM — with no spinning disks or moving parts of any kind.

SSD — How Flash Memory Works

Built from the same floating-gate transistor idea as EEPROM, organized for speed



NAND Flash Memory Chips

Cells, Pages & Blocks

Each transistor cell stores one or more bits. Cells are grouped into pages (typically 4–16 KB, the smallest unit that can be written), and pages are grouped into blocks (typically 128–256 pages, the smallest unit that can be erased).

Why You Can't Overwrite Directly

A NAND cell can only be written after it has been erased, and erasing works on a whole block at once. So updating a single page usually means writing the new data to a fresh, already-erased page and marking the old one invalid — handled automatically by the controller.

The Controller's Job

A dedicated controller chip manages wear leveling, error correction, and the mapping between the addresses the computer sees and the physical cells actually used — all invisibly and in real time.

Feature	Explanation
No Moving Parts	Data is stored and read electronically, so there is nothing to physically wear out, spin, or seek.
Very Fast	Random access to data is almost instant, since there is no read/write head that needs to physically move.
Silent & Shock-Resistant	With no motor or spinning disk, SSDs make no noise and can survive drops and vibration far better than a hard disk.
Non-Volatile	Like all secondary storage, an SSD keeps its data safely even when the power is switched off.

SSD vs Traditional Hard Disk (HDD)

Factor	HDD (Hard Disk)	SSD
Underlying technology	Spinning magnetic platters	NAND flash memory chips
Speed	Slower, especially for random access	Much faster
Moving parts	Yes — motor, platters, read/write arm	None
Noise	Audible	Silent
Price per gigabyte	Lower	Higher, though falling steadily

8 USB Flash Drives

A USB flash drive combines a small NAND flash memory chip, a controller, and a USB connector in one compact, portable package — essentially the same technology as an SSD, packaged for carrying around. Plugging it into any USB port lets it draw power and transfer data with no extra software required; it appears instantly as a removable disk on Windows, macOS, or Linux.

USB Standard	Approx. Speed	Notes
USB 2.0	Up to ~60 MB/s	Older, still common on budget drives
USB 3.0 / 3.1 / 3.2	Several hundred MB/s to a few GB/s	Much faster; look for the blue connector or “SS” logo

Common uses: transferring files between computers, creating a bootable operating system installer, running portable applications without installing them, and making quick backups of important documents.

9 Cloud Storage — The Basics

Cloud storage means storing your data on remote servers, owned and maintained by a third-party provider, instead of on your own device — and accessing that data over the internet from anywhere, on any device.

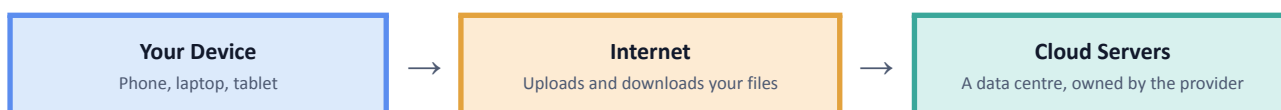


Fig 4.2 — How cloud storage works: your files travel over the internet to a remote data centre.

Common examples: Google Drive, Microsoft OneDrive, Dropbox, Apple iCloud, and Amazon Drive.

Advantages of Cloud Storage

- Access from anywhere — your files are available on any device with an internet connection, not just the one you saved them on.
- Automatic backup — if your laptop or phone is lost, damaged, or stolen, your files are still safe on the provider's servers.
- Easy sharing and collaboration — multiple people can view or edit the same file without emailing copies back and forth.
- No physical device to carry — unlike a USB drive, there is nothing to lose or forget at home.

Limitations of Cloud Storage

- Requires a working internet connection — without one, your files may be completely inaccessible.
- Ongoing cost — a small amount of storage is often free, but larger capacities usually need a paid monthly or yearly subscription.
- Privacy and security concerns — your data is stored on someone else's servers, so it depends on that provider's security practices.
- Dependent on the provider — if the service has an outage, or the company shuts the service down, access to your files can be disrupted.

10 Choosing Between Local and Cloud Storage

If You Need...	Choose...
Your PC to boot and load apps fast	An SSD
To carry files between computers without internet	A USB Flash Drive
To access your files from any device, anywhere	Cloud Storage
Automatic backup in case your device is lost or damaged	Cloud Storage
To work offline with no internet available	Local storage — SSD or USB

11 Putting It All Together — A Worked Example

Consider a student writing an assignment on a laptop, from start to finish:

Step	Technology Involved	What Happens
1	ROM	The laptop's BIOS/UEFI firmware, stored in ROM, runs first and starts the boot process.
2	SSD	The operating system and the word processor application are loaded from the SSD, the laptop's main secondary storage.
3	RAM	As the student opens the word processor and starts typing, the document and program are held in RAM for fast access.
4	Cache	As the CPU repeatedly processes keystrokes and formatting, cache keeps recently used data close by, keeping everything feeling instant.
5	SSD (again)	When the student presses Save, the file is written back to the SSD, permanently.
6	USB Flash Drive	The student also copies the file onto a USB drive to hand in a physical copy at college.
7	Cloud Storage	Finally, the student uploads the file to Google Drive, so it can be opened later from the college computer, or recovered if the laptop is lost.

This single everyday task touches every technology in this lecture: ROM at start-up, RAM and cache while working, and SSD, USB, and cloud storage for saving, carrying, and backing up the final result.

12 Fill in the Blanks

- 1. _____ is a type of memory that loses its data when the power is switched off.
- 2. _____ is a type of memory that keeps its data permanently, even without power.
- 3. The two common types of RAM chips are _____ and _____.
- 4. _____ memory sits between the CPU and RAM, holding frequently used data to speed up processing.
- 5. An SSD stores data using _____ memory chips, which have no moving parts.
- 6. A USB flash drive uses the same underlying technology as a(n) _____.
- 7. Storing data on remote servers, accessed over the internet, is called _____ storage.
- 8. Two examples of cloud storage services are _____ and _____.
- 9. Cloud storage requires a working _____ connection to access your files.

- 10. The type of ROM that can be erased and rewritten electrically, without UV light, is called _____.

Answers

1. RAM | 2. ROM | 3. SRAM, DRAM | 4. Cache | 5. NAND flash | 6. SSD | 7. Cloud | 8. Google Drive, Dropbox (any two valid providers) | 9. Internet | 10. EEPROM

13 True or False

- 1. RAM is non-volatile memory.
- 2. ROM typically stores the BIOS/firmware needed to start up a computer.
- 3. Cache memory is slower than RAM.
- 4. An SSD has moving mechanical parts like a hard disk.
- 5. USB flash drives use NAND flash memory, the same technology as an SSD.
- 6. Cloud storage lets you access your files from any device with an internet connection.
- 7. Cloud storage is always completely free, regardless of how much you store.
- 8. DRAM needs to be refreshed thousands of times per second to retain its data.

Answers with Explanation

- 1. False.** RAM is volatile — it loses its data the instant power is switched off.
- 2. True.** ROM stores fixed start-up instructions, such as the BIOS or UEFI firmware, needed before an operating system can even load.
- 3. False.** Cache is much faster than RAM — that speed advantage is the entire reason cache memory exists.
- 4. False.** An SSD has no moving parts at all, unlike a hard disk drive, which relies on a spinning platter and a moving read/write head.
- 5. True.** Both SSDs and USB flash drives are built from NAND flash memory chips, descended from EEPROM.
- 6. True.** That is precisely the defining feature of cloud storage: your files are accessible from any device with an internet connection.
- 7. False.** A small amount of cloud storage is often free, but larger capacities usually require a paid subscription.
- 8. True.** DRAM stores each bit as a charge in a capacitor, which slowly leaks, so it must be refreshed constantly to avoid losing data.

14 Short Answer Questions (2–3 Marks)

- Differentiate between RAM and ROM. (3 marks)
- What is cache memory? Why is it used? (2 marks)
- What is an SSD? State two advantages of an SSD over a hard disk. (3 marks)
- What is a USB flash drive? Give two common uses. (2 marks)
- Define cloud storage. Give two examples. (2 marks)
- State two advantages and two limitations of cloud storage. (3 marks)
- Differentiate between local storage and cloud storage. (3 marks)
- Name the four main types of ROM. (2 marks)

15 Long Answer Model Answers (5, 10 & 15 Marks)

The following are fully written model answers. Use these as a template for structure, depth, and the level of explanation expected in your own exam answer — do not simply memorise them word for word; practise writing them in your own words using the same structure.

Q1 (5 Marks): Differentiate between RAM and ROM.

RAM (Random Access Memory) and ROM (Read Only Memory) are both forms of primary memory used directly by the CPU, but they differ in almost every other respect. RAM is volatile: it loses all its contents the instant power is switched off, which is why unsaved work disappears after a crash. It can be both read from and written to constantly, and it holds the programs and data currently in use, typically measured in gigabytes.

ROM, by contrast, is non-volatile: its contents survive completely intact even with no power at all. It is read-only, or very difficult to change, and it stores small, permanent pieces of code — such as a computer's BIOS or a printer's firmware — that are needed the moment the device is switched on, before any operating system can even be loaded. In short, RAM is fast, temporary working space; ROM is small, permanent, unchanging instruction storage.

Q2 (10 Marks): Explain how an SSD and a USB flash drive work, and compare them with a traditional hard disk.

Both an SSD (Solid State Drive) and a USB flash drive are built from NAND flash memory chips, a type of non-volatile memory descended from EEPROM. Unlike a traditional hard disk, which stores data as magnetised regions on a spinning metal platter and reads it using a physically moving arm and head, flash-based storage has no moving parts at all: data is stored and read purely electronically.

Factor	HDD (Hard Disk)	SSD / USB Flash Drive
Underlying technology	Spinning magnetic platters	NAND flash memory chips
Moving parts	Yes — motor, platters, read/write arm	None
Speed	Slower, especially for random access	Much faster (SSD); portable but slower than SSD (USB)
Durability	Sensitive to drops and vibration	Highly shock-resistant
Typical role	Bulk storage inside a desktop or laptop	SSD: main drive inside a device. USB: portable, removable storage.

An SSD is normally fitted permanently inside a computer as its main storage drive, offering speeds many times faster than a hard disk for both everyday use and starting up the computer. A USB flash drive uses the same underlying flash technology, but packaged in a small, removable case with a USB connector, trading some speed and capacity for portability — making it ideal for carrying files between computers, creating a bootable installer, or making a quick backup.

In conclusion, SSDs and USB flash drives share the same core technology and the same key advantages over a hard disk — speed, silence, and durability — but are used differently: an SSD is chosen for permanent, high-speed storage inside a device, while a USB drive is chosen for its portability.

Q3 (10 Marks): Explain cache memory and its role in bridging the speed gap between the CPU and RAM.

Modern CPUs can execute instructions far faster than RAM can supply data to them — a mismatch that would leave the CPU idle for much of the time if it always had to wait on RAM directly. Cache memory solves this by sitting physically between the CPU and RAM as a small pool of very fast memory that keeps a copy of recently and frequently used data close to the CPU core.

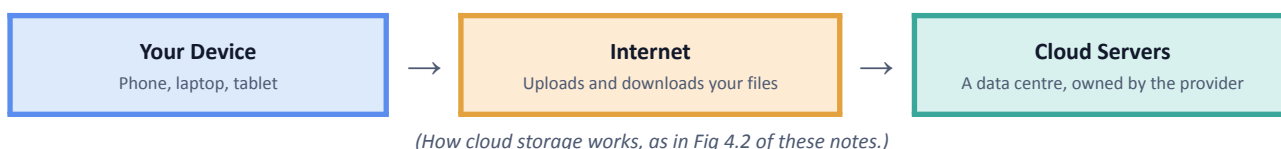
Level	Position	Typical Speed
L1	Smallest, built directly into each CPU core	Fastest
L2	Larger, still close to each core	Very fast
L3	Largest, shared across all CPU cores	Fast, but slower than L1/L2

When the CPU needs data, it checks cache first. If the data is already there, this is called a cache hit, and the data is delivered in just a few nanoseconds. If the data is not in cache, this is a cache miss, and the CPU must wait while the data is fetched from the much slower RAM — that data is then copied into cache for next time, since it is now likely to be needed again soon.

In conclusion, cache memory does not replace RAM or increase overall storage capacity; instead, it acts as a fast buffer that reduces how often the CPU has to wait for RAM, which is why a computer with a larger, well-designed cache can feel noticeably faster even with the same RAM and CPU clock speed.

Q4 (15 Marks): “Cloud storage is changing the way we think about secondary storage.” Discuss this statement by explaining what cloud storage is, how it works, its advantages and limitations, and how it compares with local storage devices like SSDs and USB drives, using suitable examples.

This statement is well justified. Traditionally, secondary storage meant a physical device — a hard disk, SSD, or USB drive — that had to be physically present with the computer to be used. Cloud storage changes this model by storing data on remote servers, owned and maintained by a third-party provider, and making it accessible over the internet from any device, anywhere.



Advantages: cloud storage services such as Google Drive, Microsoft OneDrive, and Dropbox let a user access the same files from a phone, a laptop, and a college computer, without needing to physically carry a device. Files are automatically backed up off-site, so losing or damaging a laptop does not mean losing the data, and multiple people can collaborate on the same document without emailing copies back and forth.

Limitations: cloud storage depends entirely on having a working internet connection — without one, files may be completely inaccessible. Storage beyond a small free allowance usually requires an ongoing paid subscription, and because the data lives on a third party's servers, users must trust that provider's security and reliability.

Comparison with local storage: an SSD remains the right choice for fast, offline, everyday use, since it does not depend on an internet connection and has no ongoing cost. A USB flash drive remains useful for quickly carrying files to a device with no internet access at all. Cloud storage does not replace either — rather, it adds a new, location-independent option best suited for backup, collaboration, and access across multiple devices.

In conclusion, cloud storage has not eliminated local storage, but it has changed expectations: users increasingly assume they can reach their files from anywhere and that a lost device does not mean lost data — a genuine shift in how secondary storage is thought about, even though SSDs and USB drives remain essential wherever speed, offline access, or zero ongoing cost matter most.

16 Exam Tips

- Memory trick: “RAM runs it, ROM remembers how to start, Cache keeps it close.”
- Memory trick for the ROM family: “Mask → PROM → EPROM → EEPROM” — each step makes erasing easier: never, once, UV light, electrically.

- For “differentiate” questions (RAM vs ROM, HDD vs SSD, local vs cloud storage), always answer using a table — it is faster to write and easier for the examiner to award marks clearly.
- Always name at least two real examples when asked about cloud storage providers (Google Drive, OneDrive, Dropbox, iCloud) — concrete examples earn easy marks.
- Common Mistake: Do not say cache “replaces” RAM — cache is a small, fast buffer that reduces how often the CPU needs to wait for RAM; it does not add storage capacity.
- Common Mistake: Do not say cloud storage has no cost — a small free allowance is common, but larger capacities require a paid subscription.
- For 10 and 15 mark questions, structure your answer with a clear paragraph or table per component, exactly like the model answers in Section 15 — examiners reward clear structure over long, unstructured paragraphs.

Lecture 4 — Memory & Storage — building directly on Lecture 3's Block Diagram of a Computer.